

PostgreSQL Schema — b-secure

Engine: PostgreSQL 16 + PostGIS 3.4 **Encoding:** UTF-8 **Timezone:** UTC
(application converts to IST for display) **Default PK type:** UUID
(gen_random_uuid())

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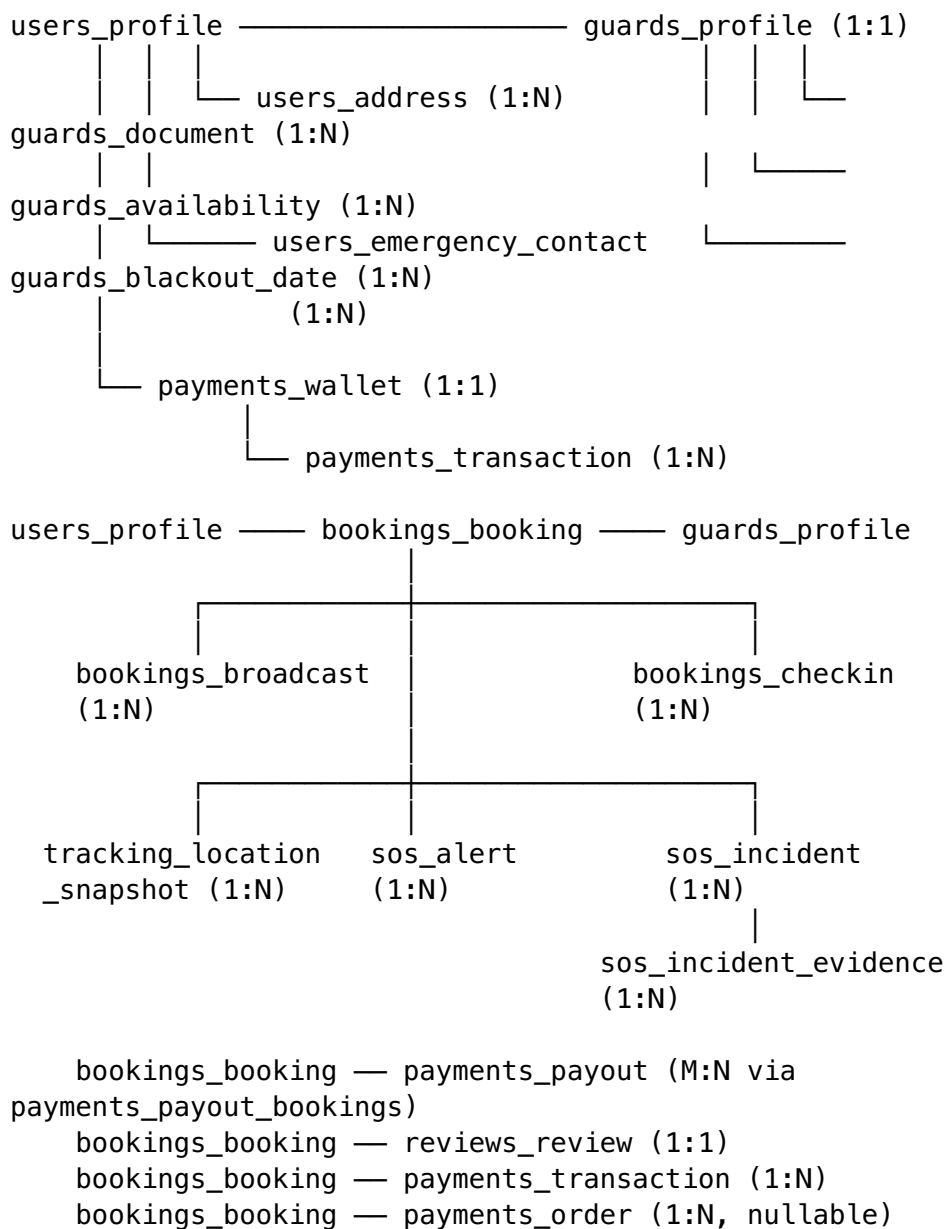
1. Extensions

```
-- Required extensions (run once on new DB)
CREATE EXTENSION IF NOT EXISTS "uuid-oss" generation; -- UUID
CREATE EXTENSION IF NOT EXISTS "postgis" types + functions; -- Geospatial
CREATE EXTENSION IF NOT EXISTS "postgis_topology" support; -- Topology
CREATE EXTENSION IF NOT EXISTS "pg_trgm" text search; -- Trigram
CREATE EXTENSION IF NOT EXISTS "btree_gist" on B-tree types (for exclusion constraints); -- GiST index
```

2. Schema Conventions

Convention	Rule	Example
Primary key	UUID, gen_random_uuid() default	id UUID PRIMARY KEY DEFAULT gen_random_uuid()
Timestamps	created_at, updated_at on every table	TIMESTAMPTZ NOT NULL DEFAULT NOW()
Soft delete	is_deleted BOOLEAN DEFAULT FALSE + deleted_at TIMESTAMPTZ	On users and guards only
Enum-style fields	VARCHAR with CHECK constraint or PostgreSQL ENUM type	status VARCHAR(20) CHECK (status IN (...))
Foreign keys	Always named {referenced_table_singular}_id	user_id UUID REFERENCES users_profile(id)
Geospatial	PostGIS GEOGRAPHY(Point, 4326) for lat/lng	current_location GEOGRAPHY(POINT, 4326)
Monetary values	NUMERIC(10,2) — never FLOAT for money	amount NUMERIC(10,2) NOT NULL
JSON fields	JSONB (not JSON) for indexable JSON	skills JSONB DEFAULT '[]'
Table names	{app}_{model} snake_case	bookings_booking, guards_profile

3. Entity Relationship Diagram



4. auth_otp_token

Stores hashed OTP codes for phone-based authentication.

```
CREATE TABLE auth_otp_token (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  phone_number VARCHAR(20)  NOT NULL,
  otp_hash    CHAR(64)      NOT NULL,
    SHA-256 hex, never plaintext
  purpose     VARCHAR(20)   NOT NULL DEFAULT 'LOGIN'
    CHECK (purpose IN
    ('LOGIN', 'PHONE_CHANGE', 'TRANSACTION')),
```

```

is_used      BOOLEAN      NOT NULL DEFAULT FALSE,
attempt_count SMALLINT    NOT NULL DEFAULT 0
              CHECK (attempt_count >= 0 AND
                    attempt_count <= 5),
expires_at   TIMESTAMPTZ  NOT NULL,
created_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW()
);

-- Indexes
CREATE INDEX idx_otp_phone_lookup
  ON auth_otp_token (phone_number, is_used, expires_at DESC);

```

Retention: Rows deleted nightly by Celery task if older than 24 hours.

5. users_profile

Central user account. Serves regular users, guards, admin, and staff roles. This is Django's AUTH_USER_MODEL.

```

CREATE TABLE users_profile (
  -- Identity
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  phone_number VARCHAR(20)  NOT NULL UNIQUE,
  email       VARCHAR(254)  UNIQUE,
  full_name   VARCHAR(150)  NOT NULL DEFAULT '',
  gender      VARCHAR(20)   DEFAULT ''
    CHECK (gender IN
      ('', 'MALE', 'FEMALE', 'OTHER', 'PREFER_NOT_TO_SAY')),
  date_of_birth DATE,
  profile_photo VARCHAR(512),
  -- S3 key

  -- Role
  role        VARCHAR(10)   NOT NULL DEFAULT 'USER'
    CHECK (role IN
      ('USER', 'GUARD', 'ADMIN', 'STAFF')),

  -- Social auth
  google_id   VARCHAR(128)  UNIQUE,
  apple_id    VARCHAR(128)  UNIQUE,

  -- Push notifications
  fcm_token   VARCHAR(512)  NOT NULL DEFAULT '',

  -- Account status
  is_active   BOOLEAN       NOT NULL DEFAULT TRUE,
  is_suspended BOOLEAN      NOT NULL DEFAULT FALSE,
  suspension_reason TEXT    NOT NULL DEFAULT '',
  suspension_ends_at TIMESTAMPTZ,
  is_deleted  BOOLEAN       NOT NULL DEFAULT FALSE,

```

```

deleted_at          TIMESTAMPTZ,

-- Django auth fields
password            VARCHAR(128) NOT NULL DEFAULT '!',           --
    Unusable for OTP users
is_staff             BOOLEAN      NOT NULL DEFAULT FALSE,
is_superuser         BOOLEAN      NOT NULL DEFAULT FALSE,
last_login           TIMESTAMPTZ,

-- Timestamps
created_at           TIMESTAMPTZ NOT NULL DEFAULT NOW(),
updated_at           TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Constraints
ALTER TABLE users_profile
    ADD CONSTRAINT users_profile_phone_e164
    CHECK (phone_number ~ '^+[1-9]\d{6,14}$');           --
    E.164 format

-- Indexes
CREATE INDEX idx_users_role_active      ON users_profile (role,
    is_active) WHERE is_deleted = FALSE;
CREATE INDEX idx_users_google_id        ON users_profile
    (google_id) WHERE google_id IS NOT NULL;
CREATE INDEX idx_users_apple_id         ON users_profile
    (apple_id) WHERE apple_id IS NOT NULL;
CREATE INDEX idx_users_fcm_token        ON users_profile
    (fcm_token) WHERE fcm_token != '';
CREATE INDEX idx_users_created_at       ON users_profile
    (created_at DESC);

```

6. users_address

Saved delivery/service addresses per user.

```

CREATE TABLE users_address (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id     UUID          NOT NULL REFERENCES
        users_profile(id) ON DELETE CASCADE,
    label       VARCHAR(20)   NOT NULL DEFAULT 'HOME'
        CHECK (label IN
            ('HOME', 'OFFICE', 'OTHER')),
    custom_label VARCHAR(50) NOT NULL DEFAULT '',
    line1       VARCHAR(255) NOT NULL,
    line2       VARCHAR(255) NOT NULL DEFAULT '',
    city        VARCHAR(100) NOT NULL,
    state       VARCHAR(100) NOT NULL,
    pincode     VARCHAR(10)  NOT NULL,
    country     VARCHAR(50)  NOT NULL DEFAULT 'India',
    latitude    NUMERIC(9,6),

```

```

        longitude    NUMERIC(9,6),
        is_default   BOOLEAN      NOT NULL DEFAULT FALSE,
        created_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
        updated_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW()
    );

-- Only one default address per user
CREATE UNIQUE INDEX idx_address_one_default
    ON users_address (user_id)
    WHERE is_default = TRUE;

CREATE INDEX idx_address_user ON users_address (user_id);

```

7. users_emergency_contact

Emergency contacts for SOS alerts. Max 5 per user (enforced in application layer).

```

CREATE TABLE users_emergency_contact (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id     UUID          NOT NULL REFERENCES
        users_profile(id) ON DELETE CASCADE,
    name        VARCHAR(150)  NOT NULL,
    phone_number VARCHAR(20)  NOT NULL,
    relationship VARCHAR(20)  NOT NULL DEFAULT 'OTHER'
        CHECK (relationship IN
            ('SPOUSE', 'PARENT', 'SIBLING', 'CHILD', 'FRIEND', 'COLLEAGUE', 'OTHER')),
    is_primary  BOOLEAN       NOT NULL DEFAULT FALSE,
    is_verified BOOLEAN       NOT NULL DEFAULT FALSE,
    created_at  TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
    updated_at  TIMESTAMPTZ   NOT NULL DEFAULT NOW()
);

-- Only one primary contact per user
CREATE UNIQUE INDEX idx_emergency_contact_one_primary
    ON users_emergency_contact (user_id)
    WHERE is_primary = TRUE;

CREATE INDEX idx_emergency_contact_user
    ON users_emergency_contact (user_id);

```

8. guards_profile

Extended profile for security guards. OneToOne with users_profile.

```

CREATE TABLE guards_profile (
    id UUID PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id UUID NOT NULL UNIQUE
        REFERENCES users_profile(id) ON DELETE CASCADE,

    -- Professional details
    guard_type VARCHAR(10) NOT NULL DEFAULT
        'UNARMED'
        CHECK (guard_type IN
            ('UNARMED', 'ARMED', 'FEMALE', 'CP0', 'EVENT', 'K9')),
    years_of_experience SMALLINT NOT NULL DEFAULT 0
        CHECK (years_of_experience >= 0),
    bio VARCHAR(500) NOT NULL DEFAULT '',
    languages_spoken JSONB NOT NULL DEFAULT '[]',
    skills JSONB NOT NULL DEFAULT '[]',

    -- Verification
    verification_status VARCHAR(25) NOT NULL DEFAULT
        'PENDING'
        CHECK
            (verification_status IN
                ('PENDING', 'UNDER_REVIEW', 'ACTIVE',
                    'SUSPENDED', 'BANNED', 'DOCUMENTS_REJECTED')),
    verified_at TIMESTAMPTZ,
    verified_by_id UUID REFERENCES
        users_profile(id) ON DELETE SET NULL,

    -- Live location (PostGIS Geography – accurate distance
    calculations)
    current_location GEOGRAPHY(POINT, 4326), --
        (lng, lat) WGS84
    last_location_update TIMESTAMPTZ,
    is_online BOOLEAN NOT NULL DEFAULT FALSE,

    -- Denormalized rating cache (updated by trigger/signal on
    new review)
    average_rating NUMERIC(3,2) NOT NULL DEFAULT 0.00
        CHECK (average_rating
            >= 0 AND average_rating <= 5),
    total_reviews INTEGER NOT NULL DEFAULT 0
        CHECK (total_reviews >= 0),
    total_sessions_completed INTEGER NOT NULL DEFAULT 0
        CHECK (total_sessions_completed >= 0),

    -- Payout details (stored encrypted at rest via RDS
    encryption)
    bank_account_number VARCHAR(20) NOT NULL DEFAULT '',
    bank_ifsc_code VARCHAR(11) NOT NULL DEFAULT '',
    upi_id VARCHAR(50) NOT NULL DEFAULT '',
    payout_preference VARCHAR(5) NOT NULL DEFAULT 'UPI'

```

```

                                CHECK
    (payout_preference IN ('BANK','UPI')),

    -- Working preferences
    preferred_work_radius_km SMALLINT NOT NULL DEFAULT 10,
    max_daily_hours          SMALLINT NOT NULL DEFAULT 12,

    created_at               TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at               TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Spatial index for proximity queries (the most critical index in
  the DB)
CREATE INDEX idx_guards_location_gist
  ON guards_profile USING GIST (current_location);

-- Partial index: only online, active guards (used by booking
  matching)
CREATE INDEX idx_guards_available
  ON guards_profile (guard_type, average_rating DESC)
  WHERE verification_status = 'ACTIVE' AND is_online = TRUE;

CREATE INDEX idx_guards_verification_status
  ON guards_profile (verification_status, created_at DESC);

```

9. guards_document

Documents uploaded for verification. One row per document type per guard.

```

CREATE TABLE guards_document (
  id          UUID PRIMARY KEY DEFAULT
    gen_random_uuid(),
  guard_id    UUID NOT NULL REFERENCES
    guards_profile(id) ON DELETE CASCADE,
  document_type VARCHAR(25) NOT NULL
    CHECK (document_type IN
      ('GOVT_ID','POLICE_CERT','PSARA_LICENSE',
      'TRAINING_CERT','ARMED_LICENSE','PROFILE_PHOTO','ADDRESS_PROOF')),
  file        VARCHAR(512) NOT NULL, -- S3
    key (private bucket)
  file_name   VARCHAR(255) NOT NULL,
  status      VARCHAR(15) NOT NULL DEFAULT 'UPLOADED'
    CHECK (status IN
      ('UPLOADED','UNDER_REVIEW','APPROVED','REJECTED','EXPIRED')),
  expiry_date DATE,
  expiry_reminder_sent BOOLEAN NOT NULL DEFAULT FALSE,

  -- Admin review

```



```

reviewed_by_id  UUID          REFERENCES users_profile(id) ON
                DELETE SET NULL,
review_notes    TEXT          NOT NULL DEFAULT '',
reviewed_at     TIMESTAMPTZ,

uploaded_at     TIMESTAMPTZ NOT NULL DEFAULT NOW(),    --
                alias for created_at
created_at      TIMESTAMPTZ NOT NULL DEFAULT NOW(),
updated_at      TIMESTAMPTZ NOT NULL DEFAULT NOW(),

UNIQUE (guard_id, document_type)
    -- one doc per type per guard
);

CREATE INDEX idx_guard_doc_status ON guards_document (status,
    created_at DESC);
CREATE INDEX idx_guard_doc_expiry ON guards_document
    (expiry_date)
    WHERE expiry_date IS NOT NULL AND status = 'APPROVED';

```

10. guards_availability

Recurring weekly availability schedule.

```

CREATE TABLE guards_availability (
    id          UUID          PRIMARY KEY DEFAULT gen_random_uuid(),
    guard_id    UUID          NOT NULL REFERENCES
        guards_profile(id) ON DELETE CASCADE,
    weekday     SMALLINT      NOT NULL CHECK (weekday BETWEEN 0 AND
        6),    -- 0=Mon, 6=Sun
    start_time  TIME          NOT NULL,
    end_time    TIME          NOT NULL,
    is_available BOOLEAN      NOT NULL DEFAULT TRUE,
    created_at  TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at  TIMESTAMPTZ NOT NULL DEFAULT NOW(),

    UNIQUE (guard_id, weekday),
    CHECK (end_time > start_time)
);

```

11. guards_blackout_date

Specific unavailable dates for a guard.

```

CREATE TABLE guards_blackout_date (
    id          UUID          PRIMARY KEY DEFAULT gen_random_uuid(),
    guard_id    UUID          NOT NULL REFERENCES guards_profile(id)
        ON DELETE CASCADE,
    date        DATE          NOT NULL,

```

```

    reason      VARCHAR(200) NOT NULL DEFAULT '',
    created_at  TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at  TIMESTAMPTZ NOT NULL DEFAULT NOW(),

    UNIQUE (guard_id, date)
);

CREATE INDEX idx_blackout_guard_date ON guards_blackout_date
    (guard_id, date);

```

12. bookings_booking

Core transaction record. Uses FSM for state management.

```

CREATE TABLE bookings_booking (
    id          UUID          PRIMARY KEY DEFAULT
                gen_random_uuid(),

    -- Participants
    user_id     UUID          NOT NULL REFERENCES
                users_profile(id) ON DELETE RESTRICT,
    guard_id    UUID          REFERENCES
                guards_profile(id) ON DELETE RESTRICT,

    -- Service details
    service_type VARCHAR(10) NOT NULL
                CHECK (service_type IN
                ('HOURLY', 'DAILY', 'WEEKLY', 'MONTHLY')),
    guard_type_requested VARCHAR(10) NOT NULL
                CHECK
                (guard_type_requested IN
                ('UNARMED', 'ARMED', 'FEMALE', 'CPO', 'EVENT')),

    -- State machine field
    status       VARCHAR(15) NOT NULL DEFAULT
                'REQUESTED'
                CHECK (status IN
                ('REQUESTED', 'BROADCAST', 'ACCEPTED', 'EN_ROUTE',
                'ARRIVED', 'ACTIVE', 'COMPLETED', 'CANCELLED',
                'DISPUTED', 'EXPIRED')),

    -- Scheduling
    scheduled_start TIMESTAMPTZ NOT NULL,
    scheduled_end   TIMESTAMPTZ NOT NULL,
    is_immediate     BOOLEAN     NOT NULL DEFAULT TRUE,

    -- Session lifecycle timestamps
    guard_accepted_at TIMESTAMPTZ,
    guard_arrived_at  TIMESTAMPTZ,

```

```

session_started_at    TIMESTAMPTZ,
session_ended_at      TIMESTAMPTZ,

-- Service location
service_address        TEXT          NOT NULL,
service_latitude       NUMERIC(9,6)  NOT NULL,
service_longitude      NUMERIC(9,6)  NOT NULL,

-- Pricing
base_rate_per_hour    NUMERIC(8,2)  NOT NULL,
surge_multiplier       NUMERIC(4,2)  NOT NULL DEFAULT 1.00,
promo_discount        NUMERIC(8,2)  NOT NULL DEFAULT 0.00,
total_amount          NUMERIC(10,2),
platform_fee          NUMERIC(8,2),
guard_earnings        NUMERIC(8,2),
tax_amount            NUMERIC(8,2),

-- OTP verification (hashed)
start_otp_hash         CHAR(64)      NOT NULL DEFAULT '',
end_otp_hash           CHAR(64)      NOT NULL DEFAULT '',

-- Cancellation
cancelled_by_id        UUID          REFERENCES
    users_profile(id) ON DELETE SET NULL,
cancellation_reason    TEXT          NOT NULL DEFAULT '',
cancelled_at           TIMESTAMPTZ,

-- Recurring
is_recurring           BOOLEAN       NOT NULL DEFAULT FALSE,
recurrence_rule        JSONB,
parent_booking_id      UUID          REFERENCES
    bookings_booking(id) ON DELETE SET NULL,

-- Invoice
invoice_s3_key         VARCHAR(512),

-- Admin
admin_notes            TEXT          NOT NULL DEFAULT '',
is_flagged             BOOLEAN       NOT NULL DEFAULT FALSE,

created_at             TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
updated_at             TIMESTAMPTZ   NOT NULL DEFAULT NOW(),

-- Constraints
CHECK (scheduled_end > scheduled_start)
);

-- Indexes (see indexes.md for full index strategy)
CREATE INDEX idx_booking_user_status ON bookings_booking
    (user_id, status, created_at DESC);
CREATE INDEX idx_booking_guard_status ON bookings_booking
    (guard_id, status) WHERE guard_id IS NOT NULL;
CREATE INDEX idx_booking_status_time ON bookings_booking
    (status, scheduled_start);

```

```

CREATE INDEX idx_booking_active          ON bookings_booking
    (user_id, guard_id, session_started_at)
    WHERE status = 'ACTIVE';
CREATE INDEX idx_booking_recurring       ON bookings_booking
    (parent_booking_id)
    WHERE is_recurring = TRUE;

```

13. bookings_broadcast

Tracks which guards received each booking request and their response.

```

CREATE TABLE bookings_broadcast (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    booking_id   UUID          NOT NULL REFERENCES
        bookings_booking(id) ON DELETE CASCADE,
    guard_id     UUID          NOT NULL REFERENCES
        guards_profile(id) ON DELETE CASCADE,
    response     VARCHAR(10)   NOT NULL DEFAULT 'SENT'
        CHECK (response IN
            ('SENT', 'ACCEPTED', 'DECLINED', 'TIMEOUT')),
    broadcast_radius_km SMALLINT NOT NULL,
    sent_at      TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
    responded_at TIMESTAMPTZ,

    UNIQUE (booking_id, guard_id)
);

CREATE INDEX idx_broadcast_booking ON bookings_broadcast
    (booking_id);
CREATE INDEX idx_broadcast_guard   ON bookings_broadcast
    (guard_id, sent_at DESC);

```

14. bookings_checkin

Guard check-ins during long active sessions (daily, weekly, monthly).

```

CREATE TABLE bookings_checkin (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    booking_id   UUID          NOT NULL REFERENCES
        bookings_booking(id) ON DELETE CASCADE,
    guard_id     UUID          NOT NULL REFERENCES
        guards_profile(id) ON DELETE CASCADE,
    latitude     NUMERIC(9,6)  NOT NULL,
    longitude    NUMERIC(9,6)  NOT NULL,
    notes        VARCHAR(200)  NOT NULL DEFAULT '',
    is_auto      BOOLEAN       NOT NULL DEFAULT FALSE,    -- TRUE =
        system auto-created
);

```

```

        created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
        updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
    );

    CREATE INDEX idx_checkin_booking_time ON bookings_checkin
        (booking_id, created_at DESC);

```

15. tracking_location_snapshot

High-volume time-series table. Guard GPS positions during active sessions.

```

CREATE TABLE tracking_location_snapshot (
    id UUID PRIMARY KEY DEFAULT
        gen_random_uuid(),
    booking_id UUID NOT NULL REFERENCES
        bookings_booking(id) ON DELETE CASCADE,
    guard_id UUID NOT NULL REFERENCES
        guards_profile(id) ON DELETE CASCADE,
    location GEOGRAPHY(POINT, 4326) NOT NULL,
        -- PostGIS point (lng, lat)
    accuracy_meters REAL,
    speed_kmh REAL,
    bearing_degrees REAL,
    timestamp TIMESTAMPTZ NOT NULL, --
        Explicit (not created_at) for replay accuracy
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Primary query pattern: all points for a session in time order
CREATE INDEX idx_location_booking_time
    ON tracking_location_snapshot (booking_id, timestamp ASC);

-- Guard history queries
CREATE INDEX idx_location_guard_time
    ON tracking_location_snapshot (guard_id, timestamp DESC);

-- Partition by month to manage growth (optional – enable when
    table > 50M rows)
-- CREATE TABLE tracking_location_snapshot_2026_01 PARTITION OF
    tracking_location_snapshot
--     FOR VALUES FROM ('2026-01-01') TO ('2026-02-01');

```

Retention policy: Rows older than 90 days deleted weekly by Celery task.

Volume: ~1 update/4 seconds per active guard = ~21,600 rows/guard/day

16. payments_wallet

One wallet per user. Balance never goes negative (enforced by CHECK and application SELECT FOR UPDATE).

```
CREATE TABLE payments_wallet (  
  id          UUID          PRIMARY KEY DEFAULT  
    gen_random_uuid(),  
  user_id     UUID          NOT NULL UNIQUE REFERENCES  
    users_profile(id) ON DELETE CASCADE,  
  balance     NUMERIC(10,2) NOT NULL DEFAULT 0.00 CHECK  
    (balance >= 0),  
  locked_balance NUMERIC(10,2) NOT NULL DEFAULT 0.00 CHECK  
    (locked_balance >= 0),  
  created_at  TIMESTAMPTZ   NOT NULL DEFAULT NOW(),  
  updated_at  TIMESTAMPTZ   NOT NULL DEFAULT NOW()  
);
```

17. payments_transaction

Immutable ledger. Rows are never updated — only inserted.

```
CREATE TABLE payments_transaction (  
  id          UUID          PRIMARY KEY DEFAULT  
    gen_random_uuid(),  
  wallet_id   UUID          NOT NULL REFERENCES  
    payments_wallet(id) ON DELETE RESTRICT,  
  transaction_type VARCHAR(20) NOT NULL  
    CHECK (transaction_type IN  
  
    ('TOPUP', 'BOOKING_DEBIT', 'BOOKING_LOCK', 'BOOKING_UNLOCK',  
  
    'REFUND', 'PROMO_CREDIT', 'REFERRAL_BONUS',  
  
    'ADMIN_CREDIT', 'ADMIN_DEBIT')),  
  amount      NUMERIC(10,2) NOT NULL CHECK (amount >  
    0),  
  balance_before NUMERIC(10,2) NOT NULL,  
  balance_after NUMERIC(10,2) NOT NULL,  
  status       VARCHAR(10)  NOT NULL DEFAULT 'PENDING'  
    CHECK (status IN  
    ('PENDING', 'SUCCESS', 'FAILED', 'REFUNDED')),  
  booking_id   UUID          REFERENCES  
    bookings_booking(id) ON DELETE SET NULL,  
  gateway      VARCHAR(10)  NOT NULL DEFAULT  
    'INTERNAL'  
    CHECK (gateway IN  
    ('RAZORPAY', 'STRIPE', 'INTERNAL')),  
  gateway_order_id VARCHAR(100) NOT NULL DEFAULT '',  
  gateway_payment_id VARCHAR(100) NOT NULL DEFAULT '',  
  gateway_signature VARCHAR(256) NOT NULL DEFAULT ''
```

```

description          VARCHAR(255) NOT NULL DEFAULT '',
admin_note           TEXT          NOT NULL DEFAULT '',
created_at           TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
updated_at           TIMESTAMPTZ   NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_txn_wallet_time ON payments_transaction
    (wallet_id, created_at DESC);
CREATE INDEX idx_txn_booking ON payments_transaction
    (booking_id, transaction_type);
CREATE INDEX idx_txn_gateway_payment ON payments_transaction
    (gateway_payment_id)
    WHERE gateway_payment_id != '';
CREATE INDEX idx_txn_status_type ON payments_transaction
    (status, transaction_type);

```

18. payments_order

Payment gateway order before payment is completed.

```

CREATE TABLE payments_order (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id     UUID          NOT NULL REFERENCES
        users_profile(id) ON DELETE RESTRICT,
    amount      NUMERIC(10,2) NOT NULL CHECK (amount > 0),
    currency    CHAR(3)       NOT NULL DEFAULT 'INR',
    purpose     VARCHAR(20)   NOT NULL DEFAULT
        'WALLET_TOPUP'
        CHECK (purpose IN
            ('WALLET_TOPUP', 'BOOKING')),
    status      VARCHAR(10)   NOT NULL DEFAULT 'CREATED'
        CHECK (status IN
            ('CREATED', 'ATTEMPTED', 'PAID', 'FAILED', 'EXPIRED')),
    gateway     VARCHAR(10)   NOT NULL CHECK (gateway IN
        ('RAZORPAY', 'STRIPE')),
    gateway_order_id VARCHAR(100) NOT NULL UNIQUE,
    gateway_response JSONB,
    booking_id   UUID          REFERENCES
        bookings_booking(id) ON DELETE SET NULL,
    created_at   TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
    updated_at   TIMESTAMPTZ   NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_order_user_status ON payments_order
    (user_id, status);
CREATE INDEX idx_order_gateway_order_id ON payments_order
    (gateway_order_id);

```

19. payments_payout

Earnings payouts to guards via bank transfer or UPI.

```
CREATE TABLE payments_payout (  
  id          UUID          PRIMARY KEY DEFAULT  
    gen_random_uuid(),  
  guard_id    UUID          NOT NULL REFERENCES  
    guards_profile(id) ON DELETE RESTRICT,  
  amount      NUMERIC(10,2) NOT NULL CHECK (amount > 0),  
  status      VARCHAR(15)   NOT NULL DEFAULT 'PENDING'  
    CHECK (status IN  
      ('PENDING', 'PROCESSING', 'COMPLETED', 'FAILED', 'ON_HOLD')),  
  period_start DATE         NOT NULL,  
  period_end   DATE         NOT NULL,  
  razorpay_payout_id VARCHAR(100) NOT NULL DEFAULT '',  
  bank_reference VARCHAR(100) NOT NULL DEFAULT '',  
  failure_reason TEXT       NOT NULL DEFAULT '',  
  processed_at TIMESTAMPTZ,  
  processed_by_id UUID       REFERENCES  
    users_profile(id) ON DELETE SET NULL,  
  created_at   TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  updated_at   TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  
  CHECK (period_end >= period_start),  
  UNIQUE (guard_id, period_start, period_end)      -- Prevent  
    duplicate payouts for same period  
);  
  
CREATE INDEX idx_payout_guard_status ON payments_payout  
  (guard_id, status);  
CREATE INDEX idx_payout_status_period ON payments_payout (status,  
  period_start);
```

20. payments_payout_bookings (M2M)

Junction table linking payouts to the bookings they cover.

```
CREATE TABLE payments_payout_bookings (  
  id          BIGSERIAL PRIMARY KEY,  
  payout_id   UUID       NOT NULL REFERENCES  
    payments_payout(id) ON DELETE CASCADE,  
  booking_id  UUID       NOT NULL REFERENCES  
    bookings_booking(id) ON DELETE CASCADE,  
  
  UNIQUE (payout_id, booking_id)  
);
```



```
CREATE INDEX idx_payout_booking_payout ON
payments_payout_bookings (payout_id);
CREATE INDEX idx_payout_booking_booking ON
payments_payout_bookings (booking_id);
```

21. notifications_log

Full audit trail of every notification sent.

```
CREATE TABLE notifications_log (
  id UUID PRIMARY KEY DEFAULT
    gen_random_uuid(),
  recipient_id UUID NOT NULL REFERENCES
    users_profile(id) ON DELETE CASCADE,
  channel VARCHAR(10) NOT NULL
    CHECK (channel IN
      ('PUSH', 'SMS', 'EMAIL', 'IN_APP', 'WHATSAPP')),
  notification_type VARCHAR(50) NOT NULL,
  title VARCHAR(255) NOT NULL DEFAULT '',
  body TEXT NOT NULL,
  data JSONB NOT NULL DEFAULT '{}',
  status VARCHAR(10) NOT NULL DEFAULT 'QUEUED'
    CHECK (status IN
      ('QUEUED', 'SENT', 'DELIVERED', 'FAILED', 'BOUNCED')),
  provider_message_id VARCHAR(255) NOT NULL DEFAULT '',
  failure_reason TEXT NOT NULL DEFAULT '',
  is_read BOOLEAN NOT NULL DEFAULT FALSE,
  read_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

-- Unread in-app notifications (badge count + inbox query)

```
CREATE INDEX idx_notif_unread
ON notifications_log (recipient_id, created_at DESC)
WHERE is_read = FALSE AND channel = 'IN_APP';
```

```
CREATE INDEX idx_notif_type_status ON notifications_log
(notification_type, status);
```

22. notifications_preference

User opt-in/out settings per notification channel.

```
CREATE TABLE notifications_preference (
  id UUID PRIMARY KEY DEFAULT
    gen_random_uuid(),
  user_id UUID NOT NULL UNIQUE REFERENCES
    users_profile(id) ON DELETE CASCADE,
```

```

push_enabled      BOOLEAN NOT NULL DEFAULT TRUE,
sms_enabled       BOOLEAN NOT NULL DEFAULT TRUE,
email_enabled     BOOLEAN NOT NULL DEFAULT TRUE,
marketing_push    BOOLEAN NOT NULL DEFAULT FALSE,
marketing_email   BOOLEAN NOT NULL DEFAULT FALSE,
created_at        TIMESTAMPTZ NOT NULL DEFAULT NOW(),
updated_at        TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

```

23. sos_alert

Mission-critical SOS events. Writes are always synchronous.

```

CREATE TABLE sos_alert (
  id                UUID                PRIMARY KEY DEFAULT
    gen_random_uuid(),
  user_id           UUID                NOT NULL REFERENCES
    users_profile(id) ON DELETE RESTRICT,
  booking_id        UUID                REFERENCES
    bookings_booking(id) ON DELETE SET NULL,
  trigger_method     VARCHAR(20)        NOT NULL
    CHECK (trigger_method IN
      ('BUTTON', 'SHAKE', 'AUTO_CHECKIN',
      'GUARD_OFFLINE', 'GUARD_DISTRESS')),
  status            VARCHAR(15)         NOT NULL DEFAULT
    'TRIGGERED'
    CHECK (status IN
      ('TRIGGERED', 'ACKNOWLEDGED', 'RESPONDING',
      'RESOLVED', 'FALSE_ALARM')),

  -- Location at time of trigger
  latitude           NUMERIC(9,6)        NOT NULL,
  longitude          NUMERIC(9,6)        NOT NULL,
  location_accuracy_meters REAL,

  -- Response tracking
  assigned_to_id     UUID                REFERENCES
    users_profile(id) ON DELETE SET NULL,
  acknowledged_at    TIMESTAMPTZ,
  resolved_at        TIMESTAMPTZ,
  resolution_notes    TEXT                NOT NULL DEFAULT '',

  -- Optional audio recording
  recording_file      VARCHAR(512),      --
    S3 key

  created_at         TIMESTAMPTZ        NOT NULL DEFAULT NOW(),
  updated_at         TIMESTAMPTZ        NOT NULL DEFAULT NOW()
);

```

```
);

-- Open alerts only (small set, always fast)
CREATE INDEX idx_sos_open
  ON sos_alert (created_at DESC)
  WHERE status NOT IN ('RESOLVED', 'FALSE_ALARM');

CREATE INDEX idx_sos_user    ON sos_alert (user_id, created_at
  DESC);
CREATE INDEX idx_sos_booking ON sos_alert (booking_id) WHERE
  booking_id IS NOT NULL;
```

24. sos_emergency_contact_alert

Tracks which emergency contacts were notified per SOS.

```
CREATE TABLE sos_emergency_contact_alert (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  sos_alert_id UUID          NOT NULL REFERENCES sos_alert(id)
    ON DELETE CASCADE,
  contact_name VARCHAR(150) NOT NULL,
  contact_phone VARCHAR(20) NOT NULL,
  sms_sent      BOOLEAN      NOT NULL DEFAULT FALSE,
  sms_delivered BOOLEAN      NOT NULL DEFAULT FALSE,
  call_attempted BOOLEAN      NOT NULL DEFAULT FALSE,
  created_at    TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
  updated_at    TIMESTAMPTZ  NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_sos_contact_sos ON sos_emergency_contact_alert
  (sos_alert_id);
```

25. sos_incident

User or guard filed incident reports (retrospective, separate from SOS).

```
CREATE TABLE sos_incident (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  booking_id   UUID          NOT NULL REFERENCES
    bookings_booking(id) ON DELETE RESTRICT,
  filed_by_id  UUID          NOT NULL REFERENCES
    users_profile(id) ON DELETE RESTRICT,
  incident_type VARCHAR(30)  NOT NULL
    CHECK (incident_type IN
      ('GUARD_MISCONDUCT', 'GUARD_NO_SHOW', 'GUARD_EARLY_DEPARTURE',
```

```

        'THREATENING_BEHAVIOUR', 'THEFT', 'PROPERTY_DAMAGE',
        'DANGEROUS_CLIENT', 'SAFETY_CONCERN', 'OTHER')),
severity      VARCHAR(10) NOT NULL DEFAULT 'MEDIUM'
              CHECK (severity IN
                ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')),
description   TEXT      NOT NULL,
status        VARCHAR(15) NOT NULL DEFAULT 'OPEN'
              CHECK (status IN
                ('OPEN', 'IN_REVIEW', 'RESOLVED', 'CLOSED')),
assigned_to_id UUID      REFERENCES users_profile(id) ON
              DELETE SET NULL,
resolution_notes TEXT    NOT NULL DEFAULT '',
resolved_at   TIMESTAMPTZ,
created_at    TIMESTAMPTZ NOT NULL DEFAULT NOW(),
updated_at    TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_incident_status ON sos_incident (status,
created_at DESC);
CREATE INDEX idx_incident_booking ON sos_incident (booking_id);
CREATE INDEX idx_incident_severity ON sos_incident (severity,
status);

```

26. sos_incident_evidence

Photo/video evidence attached to incidents.

```

CREATE TABLE sos_incident_evidence (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  incident_id UUID          NOT NULL REFERENCES sos_incident(id)
    ON DELETE CASCADE,
  file        VARCHAR(512) NOT NULL,                -- S3 key
  file_type   VARCHAR(10) NOT NULL CHECK (file_type IN
    ('IMAGE', 'VIDEO')),
  description VARCHAR(255) NOT NULL DEFAULT '',
  created_at  TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_at  TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_evidence_incident ON sos_incident_evidence
  (incident_id);

```

27. reviews_review

Post-session ratings. One review per completed booking.

```

CREATE TABLE reviews_review (
    id UUID PRIMARY KEY DEFAULT
        gen_random_uuid(),
    booking_id UUID NOT NULL UNIQUE REFERENCES
        bookings_booking(id) ON DELETE RESTRICT,
    reviewer_id UUID NOT NULL REFERENCES
        users_profile(id) ON DELETE RESTRICT,
    guard_id UUID NOT NULL REFERENCES
        guards_profile(id) ON DELETE RESTRICT,

    -- Ratings 1-5
    overall_rating SMALLINT NOT NULL CHECK
        (overall_rating BETWEEN 1 AND 5),
    punctuality_rating SMALLINT CHECK (punctuality_rating
        BETWEEN 1 AND 5),
    professionalism_rating SMALLINT CHECK
        (professionalism_rating BETWEEN 1 AND 5),
    communication_rating SMALLINT CHECK (communication_rating
        BETWEEN 1 AND 5),
    alertness_rating SMALLINT CHECK (alertness_rating
        BETWEEN 1 AND 5),

    comment VARCHAR(1000) NOT NULL DEFAULT '',
    is_flagged BOOLEAN NOT NULL DEFAULT FALSE,
    flag_reason VARCHAR(255) NOT NULL DEFAULT '',
    is_hidden BOOLEAN NOT NULL DEFAULT FALSE,

    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_review_guard ON reviews_review (guard_id,
    overall_rating DESC, created_at DESC);
CREATE INDEX idx_review_reviewer ON reviews_review
    (reviewer_id);
CREATE INDEX idx_review_flagged ON reviews_review
    (is_flagged) WHERE is_flagged = TRUE;

```

Trigger to update guard's average_rating after insert/update:

```

CREATE OR REPLACE FUNCTION update_guard_average_rating()
RETURNS TRIGGER AS $$
BEGIN
    UPDATE guards_profile
    SET
        average_rating = (
            SELECT COALESCE(AVG(overall_rating::NUMERIC), 0)
            FROM reviews_review
            WHERE guard_id = NEW.guard_id AND is_hidden = FALSE
        ),
        total_reviews = (
            SELECT COUNT(*)
            FROM reviews_review
            WHERE guard_id = NEW.guard_id AND is_hidden = FALSE
        )

```

```

    WHERE id = NEW.guard_id;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_update_guard_rating
    AFTER INSERT OR UPDATE ON reviews_review
    FOR EACH ROW EXECUTE FUNCTION update_guard_average_rating();

```

28. analytics_daily_stats

Pre-aggregated daily metrics for instant admin dashboard reads.

```

CREATE TABLE analytics_daily_stats (
    id                BIGSERIAL          PRIMARY KEY,
    date              DATE                NOT NULL UNIQUE,

    -- Bookings
    total_bookings    INTEGER            NOT NULL DEFAULT 0,
    completed_bookings INTEGER          NOT NULL DEFAULT 0,
    cancelled_bookings INTEGER          NOT NULL DEFAULT 0,
    disputed_bookings INTEGER          NOT NULL DEFAULT 0,
    hourly_bookings  INTEGER          NOT NULL DEFAULT 0,
    daily_bookings    INTEGER          NOT NULL DEFAULT 0,
    weekly_bookings   INTEGER          NOT NULL DEFAULT 0,
    monthly_bookings  INTEGER          NOT NULL DEFAULT 0,

    -- Revenue
    gross_revenue      NUMERIC(12,2) NOT NULL DEFAULT 0,
    platform_fees_collected NUMERIC(12,2) NOT NULL DEFAULT 0,
    guard_earnings_paid NUMERIC(12,2) NOT NULL DEFAULT 0,
    refunds_issued     NUMERIC(12,2) NOT NULL DEFAULT 0,

    -- Users & Guards
    new_users          INTEGER          NOT NULL DEFAULT 0,
    active_users       INTEGER          NOT NULL DEFAULT 0,
    new_guards         INTEGER          NOT NULL DEFAULT 0,
    active_guards      INTEGER          NOT NULL DEFAULT 0,

    -- Safety
    sos_alerts         INTEGER          NOT NULL DEFAULT 0,
    incidents_filed    INTEGER          NOT NULL DEFAULT 0,

    created_at         TIMESTAMPTZ     NOT NULL DEFAULT NOW(),
    updated_at         TIMESTAMPTZ     NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_daily_stats_date ON analytics_daily_stats (date
DESC);

```

29. Django System Tables

Django creates these automatically. Documented here for completeness.

Table	Purpose
<code>django_migrations</code>	Migration history — which migrations have been applied
<code>django_content_type</code>	Content type framework — used by permissions
<code>auth_permission</code>	Permission definitions
<code>auth_group</code>	User groups
<code>auth_group_permissions</code>	M2M: groups ↗ permissions
<code>users_profile_groups</code>	M2M: users ↗ groups
<code>users_profile_user_permissions</code>	M2M: users ↗ direct permissions
<code>django_session</code>	Server-side sessions (if not using JWT-only)
<code>django_admin_log</code>	Django admin action log
<code>token_blacklist_outstandingtoken</code>	JWT outstanding tokens (SimpleJWT)
<code>token_blacklist_blacklistedtoken</code>	Blacklisted refresh tokens (SimpleJWT)
<code>django_celery_beat_periodictask</code>	Celery beat schedule (stored in DB)
<code>django_celery_beat_crontabschedule</code>	Crontab schedule definitions
<code>django_celery_results_taskresult</code>	Celery task result storage